

NMEA Link – User Manual



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Introduction

The **NMEA Link** is a portable diagnostic and test device designed for the monitoring and generation of **NMEA 0183** serial data used in marine navigation and communication systems.

It is intended for technicians and engineers performing commissioning, troubleshooting, validation, and bench testing of marine electronic equipment.

The device operates fully standalone and does not require any external software. All interaction is performed through a built-in Wi-Fi captive portal using a standard web browser.

The NMEA Link was developed within marine electronics projects at **THEMYS**, combining real-world operational experience with laboratory testing.

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1 Device Overview

The NMEA Link acts as a diagnostic interface between an NMEA 0183 data source and the user, enabling real-time visualization of NMEA traffic and controlled sentence generation.

The device combines:

- Differential serial interface (A/B) compatible with NMEA 0183,
- External probe with DB9 connector and test leads for direct connection,
- Local OLED display and visual status indicators,
- Wi-Fi captive portal for monitoring and control,
- Internal battery for fully standalone operation.

The device is intended strictly as a **diagnostic and test tool**. It does **not** configure, modify, or interfere with the normal operation of the connected equipment.



Figure 1: NMEA Link device.

To acquire NMEA 0183 signals from external equipment, the NMEA Link uses an external probe connected through a **DB9 connector**.

The probe provides direct access to the differential NMEA 0183 signal pair, commonly identified as:

- **A / B** (differential signal),
- or **+ / -** depending on manufacturer terminology.

The probe is intended for diagnostic purposes and allows technicians to connect directly to the NMEA signal lines of the equipment under test without modifying the original installation.

NMEA 0183 is a differential serial interface. Correct polarity (A/B or +/-) must be respected to ensure proper data reception.



Figure 2: External probe with DB9 connector used to acquire NMEA 0183 A/B (or +/-) signals.

The external probe is designed to interface with differential serial signals commonly used in marine electronics.

NMEA 0183 is electrically compatible with **RS-422** and, in some installations, with **RS-485** signaling. These standards use a differential A/B (or +/-) pair, which improves noise immunity and ensures reliable data acquisition in marine environments.

For this reason, the probe provides two signal leads intended to be connected directly to the differential NMEA 0183 output of the equipment under test.

2 Powering the Device

The device includes an internal 3.7 V Li-ion 18650 battery and can be powered either internally or via USB.



Figure 3: Internal 18650 Li-ion battery.

The device supports:

- Standalone battery operation,
- Operation while connected to a USB charger,
- Operation while connected to a USB power bank.

When external power is applied, the device continues operating normally while charging the battery.



Figure 4: USB charger and power bank used to power and charge the device.

The USB connector is used exclusively for power and battery charging. It does not provide any data interface.

3 Quick Start

3.1 Step 1 – Power On

Use the ON/OFF switch to power on the device.

For safety reasons, the device always starts in PAUSE mode. No NMEA data is processed or transmitted until Start is pressed from the web interface.

After powering on the device, the OLED display turns on immediately and provides essential information to the user. The OLED display indicates:

- The current operating mode (**RUN** or **PAUSE**),
- General device status,
- The **automatically detected sensor type** based on the incoming NMEA 0183 sentences.

The sensor detection is fully automatic and does not require any user configuration. If different NMEA sources are connected, the displayed sensor may change dynamically.



Figure 5: OLED display active after power-on, showing operating mode and baudrate.

3.2 Step 2 – Wi-Fi Connection

Connect to the Wi-Fi network:

- SSID: NMEA_Link
- Password: 12345678

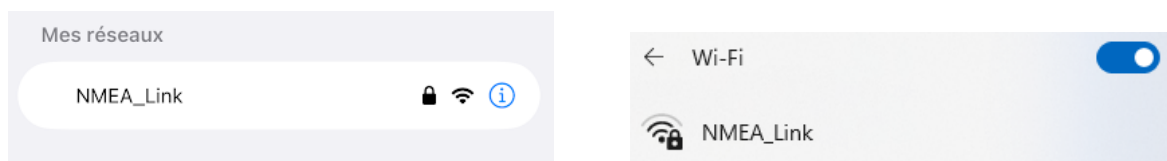


Figure 6: Wi-Fi connection to the NMEA Link using a mobile device and a PC.

3.3 Step 3 – Web Interface Access

Once connected to the NMEA Link Wi-Fi network, the captive portal **opens automatically** on most devices.

If the captive portal does not open automatically, open a web browser and manually navigate to:

`http://192.168.4.1`

This address can always be used to access the web interface.

4 User Interface – Captive Portal

4.1 Main Menu

The main menu provides access to the Monitor, Generator, and OTA update pages.

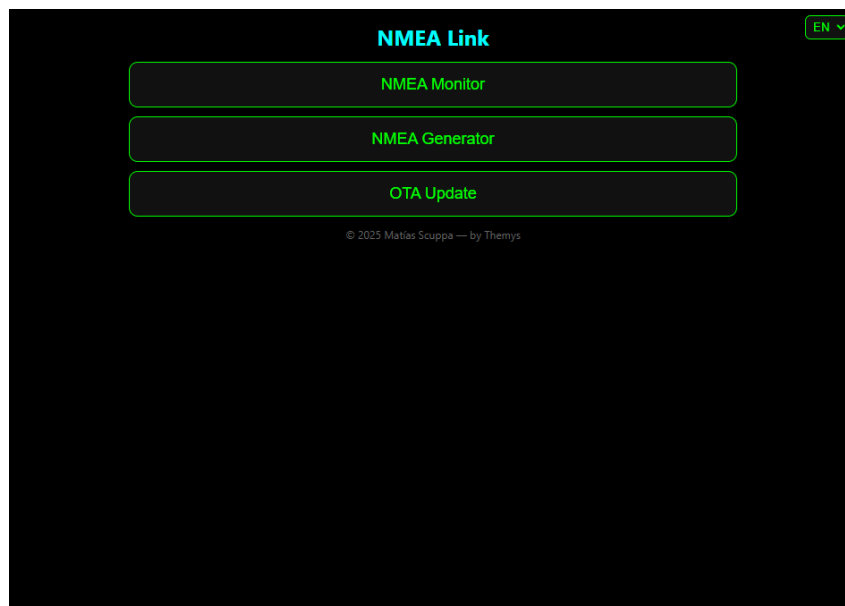


Figure 7: Captive portal main menu.

4.2 NMEA Monitor

The NMEA Monitor provides real-time visualization of incoming NMEA 0183 sentences received from the connected equipment.

In addition to sentence classification, the NMEA Link performs **checksum verification** on all received NMEA 0183 sentences.

Only sentences with a valid checksum are considered correct and processed by the system. This mechanism ensures that corrupted or incomplete data caused by noise, wiring issues, or transmission errors can be detected during troubleshooting and commissioning activities.

The device **automatically analyzes each received sentence** to determine the originating sensor type.

Based on this analysis:

- Sentences are classified by sensor,
- The detected sensor is highlighted in the Monitor interface,
- The active sensor is also reflected on the OLED display.



Figure 8: NMEA Monitor mode displaying live NMEA traffic with automatic sensor classification.

4.2.1 Serial Viewer and Color Coding

Incoming sentences are displayed in a scrolling serial viewer and color-coded according to sensor category.

4.2.2 Sensor Categories

Supported categories include GPS, AIS, Sounder, Velocity, Heading, Radar, Weather, Transducer, and Other.

4.2.3 Filters and Controls

The Monitor page provides:

- Sensor-based filters,
- Manual baudrate selection,
- Start / Pause control,
- Clear viewer function,
- Display speed adjustment.

4.2.4 Supported Baud Rates

- 4800
- 9600
- 38400
- 115200

4.2.5 Operating Behavior

Leaving the page or returning to the main menu automatically pauses data processing.

4.3 NMEA Generator

The NMEA Generator allows controlled transmission of valid NMEA 0183 sentences for interface testing and system validation.



The screenshot displays the NMEA Generator configuration interface. It features a dark theme with red and blue highlights. At the top, there are two dropdown menus: 'Sensor' (set to 'CUSTOM') and 'Sentence type' (set to 'CUSTOM'). Below these is a text input field containing '\$GPCUS.FIELD1.FIELD2'. The 'Interval' section has four buttons: '0.1s', '0.5s' (highlighted in red), '1s', and '2s'. The 'Baudrate' section has four buttons: '4800' (highlighted in red), '9600', '38400', and '115200'. A large empty text area is positioned below the baudrate buttons. At the bottom, there are two buttons: 'Start' (with a play icon) and 'Clear' (with a trash icon). Below these are two more buttons: 'NMEA Monitor' and 'Main Menu' (with a house icon). The footer text reads '© 2023 Matias Scuppa — by Themys'.

Figure 9: NMEA Generator mode used to transmit test sentences.

4.3.1 Generator Modes

The generator supports:

- Predefined sentences associated with common sensor types,
- A custom sentence input field.

4.3.2 Controls

The Generator page provides:

- Baudrate selection,
- Transmission interval selection,
- Start / Pause control,
- Transmission viewer.

4.4 Update (OTA)

Firmware updates are performed through the captive portal by uploading a BIN file.

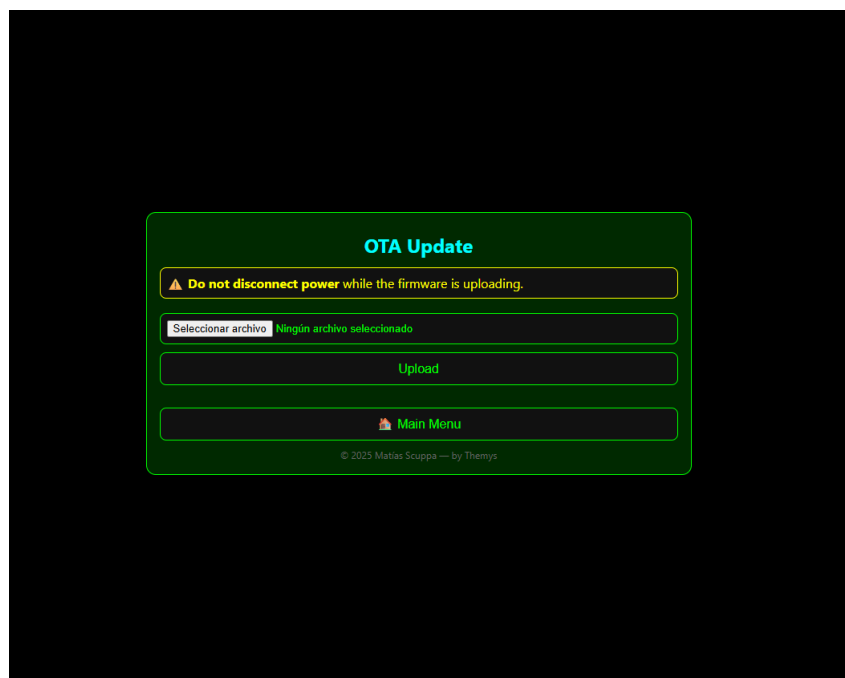


Figure 10: Update (OTA) page used for firmware upgrades.

After a successful update, the device automatically restarts.

- Do not disconnect power during firmware upload.

5 Troubleshooting

This section provides guidance for resolving common operational issues related to power, Wi-Fi connectivity, NMEA data reception, generator output, and OTA updates.

Device does not power ON

Check battery charge, USB power source, and ON/OFF switch position.

Wi-Fi network not visible

Ensure the device is powered ON and wait a few seconds for Wi-Fi initialization.

Captive portal does not open

Open a browser manually and navigate to `http://192.168.4.1`.

No NMEA sentences displayed

Verify wiring, baudrate selection, and that the connected equipment is transmitting data. If no data is displayed and the source is known to be active, check the polarity of the NMEA 0183 differential pair.

- Swap the A/B or +/– connections of the external probe.
- Ensure correct identification of the NMEA output terminals on the equipment under test.

Reversed A/B (or +/–) connections will prevent correct data reception but do not damage the NMEA Link or the connected equipment.

Generator does not transmit

Check RUN/PAUSE state, baudrate compatibility, and wiring.

OTA update fails

Verify the firmware file, ensure stable power, and retry the update.

6 Safety and Usage Notes

The NMEA Link is intended strictly as a **diagnostic and test device** for marine electronic systems.

The following usage guidelines must be observed:

- The device must not be used as a permanent part of the vessel navigation system.
- The NMEA Link must not replace certified navigation or safety equipment.
- The device is intended for commissioning, troubleshooting, validation, and bench testing only.
- Connections must be made exclusively to standard NMEA 0183 signal lines.
- The device must not be connected to energized power lines outside the NMEA 0183 specification.

Failure to follow these guidelines may result in incorrect system behavior or damage to connected equipment.

7 Typical Use Cases

The NMEA Link is designed to support a wide range of diagnostic and testing scenarios commonly encountered in marine electronics installations.

Typical use cases include:

- Commissioning and validation of GPS, AIS, gyro, log, sounder, and weather sensors.
- Verification of NMEA 0183 output from navigation and communication equipment.
- Troubleshooting intermittent or incorrect NMEA data on board.
- Bench testing of marine electronic equipment prior to installation.
- Validation of system integration between multiple NMEA data sources.

The automatic sensor detection and classification features allow technicians to quickly identify active data sources without prior configuration.

8 Future Developments

The NMEA Link platform has been designed with future expansion and certification requirements in mind. Several firmware and system-level improvements are currently under evaluation as part of the product roadmap.

Planned and ongoing development areas include:

- **Automatic baudrate detection:** Implementation of reliable baudrate auto-detection algorithms to allow seamless connection to unknown NMEA 0183 data sources without manual configuration.
- **Improved robustness and validation mechanisms:** Enhanced sentence validation, checksum handling, and error detection to improve reliability during noisy or unstable signal conditions.
- **Extended internal diagnostics:** Additional internal status monitoring intended to support fault analysis and long-term reliability assessments.
- **Firmware architecture refinement:** Continued modularization of the firmware to facilitate verification, traceability, and controlled evolution of features.
- **Certification-oriented development:** Alignment of hardware and firmware design practices with the requirements of marine classification societies such as **RINA**, with the goal of enabling future product certification.

As part of its long-term product roadmap, the NMEA Link is being developed with future certification and compliance objectives in mind.

Potential certification paths under evaluation include alignment with marine classification societies such as RINA, Bureau Veritas, and DNV, depending on the intended scope of use and deployment.

In parallel, the design and development process considers applicable regulatory and normative frameworks, including European conformity requirements such as CE marking and relevant IEC standards for marine electronic equipment.

These considerations aim to ensure that future evolutions of the NMEA Link maintain a high level of reliability, traceability, and compliance while preserving its primary role as a dedicated diagnostic and test instrument.

These developments are intended to further improve usability, reliability, and compliance while preserving the NMEA Link's role as a dedicated diagnostic and test instrument.

9 Development Status Disclaimer

The NMEA Link is currently under active development.

Firmware, hardware, and visual or mechanical aspects of the device may be subject to change as part of ongoing improvement efforts.

The information provided in this manual reflects the current state of the product at the time of publication and may be updated without prior notice.

10 Author's Note

The NMEA Link was developed based on practical experience in marine electronics installation, commissioning, and troubleshooting.

The design philosophy behind this device focuses on simplicity, reliability, and real-world usability, aiming to provide technicians and engineers with a practical diagnostic tool that reflects actual operational needs encountered in the field.

This user manual is intended to evolve alongside the product, incorporating feedback from real installations and continuous technical improvements.

11 Revision History

- **Rev 1.0** – Initial release of the NMEA Link User Manual.